

Mobile Radio Propagation

There are two basic ways of transmitting an electro-magnetic (EM) signal, through a guided medium or through an unguided medium. Guided mediums such as coaxial cables and fiber optic cables are far less hostile toward the information carrying EM signal than the wireless or the unguided medium. It presents challenges and conditions which are unique for this kind of transmissions. A signal, as it travels through the wireless channel, undergoes many kinds of propagation effects such as reflection, diffraction and scattering, due to the presence of buildings, mountains and other such obstructions. Reflection occurs when the EM waves impinge on objects which are much greater than the wavelength of the traveling wave. Diffraction is a phenomena occurring when the wave interacts with a surface having sharp irregularities. Scattering occurs when the medium through the wave is traveling contains objects which are much smaller than the wavelength of the EM wave. These varied phenomena's lead to large scale and small scale propagation losses. Due to the inherent randomness associated with such channels they are best described with the help of statistical models. Models which predict the mean signal strength for arbitrary transmitter receiver distances are termed as large scale propagation models. These are termed so because they predict the average signal strength for large Tx-Rx separations, typically for hundreds of kilometers.

Free Space Propagation Model

Although EM signals when traveling through wireless channels experience fading effects due to various effects, but in some cases the transmission is with a direct line of sight such as in satellite communication. Free space model predicts that the received power decays as negative square root of the distance. Friis free space equation is given by

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

where P_t is the transmitted power, $P_r(d)$ is the received power, G_t is the transmitter antenna gain, G_r is the receiver antenna gain, d is the Tx-Rx separation and L is the system loss factor depend upon line attenuation, filter losses and antenna losses and not related to propagation. The gain of the antenna is related to the effective aperture of the antenna which in turn is dependent upon the physical size of the antenna as given below

$$G = 4\pi A_e / \lambda^2.$$

The path loss, representing the attenuation suffered by the signal as it travels through the wireless channel is given by the difference of the transmitted and received power in dB and is expressed as:

$$PL (dB) = 10 \log \frac{P_t}{P_r} = -10 \log \left[\frac{\lambda^2}{(4\pi d)^2} \right]$$

The fields of an antenna can broadly be classified in two regions, the far field and the near field. It is in the far field that the propagating waves act as plane waves and the power decays inversely with distance. The far field region is also termed as Fraunhofer region and the Friis equation holds in this region. Hence, the Friis equation is used only beyond the far field distance, d_f , which is dependent upon the largest dimension of the antenna as

$$d_f = 2D^2/\lambda.$$

Also we can see that the Friis equation is not defined for $d=0$. For this reason, we use a close in distance, d_o , as a reference point. The power received, $P_r(d)$, is then given by:

$$P_r(d) = P_r(d_o)(d_o/d)^2$$

Basic Methods of Propagation

Reflection, diffraction and scattering are the three fundamental phenomena that cause signal propagation in a mobile communication system, apart from LoS communication. The most important parameter, predicted by propagation models based on above three phenomena, is the received power. The physics of the above phenomena may also be used to describe small scale fading and multipath propagation. The following subsections give an outline of these phenomena.

Reflection

Reflection occurs when an electromagnetic wave falls on an object, which has very large dimensions as compared to the wavelength of the propagating wave. For example, such objects can be the earth, buildings and walls. When a radio wave falls on another medium having different electrical properties, a part of it is transmitted into it, while some energy is reflected back. Let us see some special cases. If the medium on which the e.m. wave is incident is a dielectric, some energy is reflected back and some energy is transmitted. If the medium is a perfect conductor, all energy is reflected back to the first medium. The amount of energy that is reflected back depends on the polarization of the e.m. wave. Another particular case of interest arises in parallel polarization, when no reflection occurs in the medium of origin. This would occur, when the incident angle would be such that the reflection coefficient is equal to zero. This angle is the Brewster's angle.

Diffraction

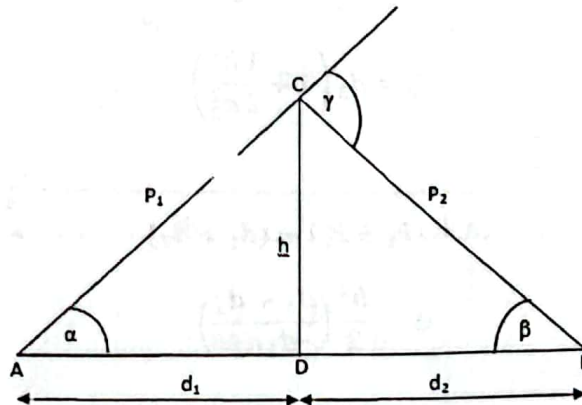
Diffraction is a phenomenon that takes place when the radio wave strikes a surface and changes its direction of propagation owing to the inability of the surface to absorb it. The loss due to diffraction depends upon the kind of obstruction in the path. In practice, the mobile antenna is at a much lower height than the base station antenna, and there may be high buildings or hills in the area. Thus, the signal undergoes diffraction in reaching the mobile antenna. This phenomenon is also known as 'shadowing' because the mobile receiver is in the shadow of these structures.

Also, diffraction is the phenomenon due to which an EM wave can propagate beyond the horizon, around the curved earth's surface and obstructions like tall buildings. As the user moves deeper into the shadowed region, the received field strength decreases. But the diffraction field still exists and it has enough strength to yield a good signal. This phenomenon can be explained by the Huygen's principle, according to which, every point on a wavefront acts as point sources

for the production of secondary wavelets, and they combine to produce a new wavefront in the direction of propagation. The propagation of secondary wavelets in the shadowed region results in diffraction. The field in the shadowed region is the vector sum of the electric field components of all the secondary wavelets that are received by the receiver.

Fresnel Zone Geometry:

Consider a transmitter and receiver located across an obstruction as shown in the diagram.



Assumptions:

- $d_1, d_2 \gg h$
- A : Transmitter
- B : Receiver
- CD : Height of obstruction above LOS = h
- C : Diffraction point
- P_1 and P_2 : Path traversed by EMW
- α, β , and γ : various angle made by EMW while travelling from TX to RX
- d_1 and d_2 : distance of obstruction from transmitter and receiver respectively

In the triangle ACD,

$$P_1 = \sqrt{d_1^2 + h^2}$$

$$P_1 = \sqrt{1 + \left(\frac{h}{d_1}\right)^2}$$

Applying Taylor's series,

$$P_1 = d_1 \left(1 + \frac{1}{2} \frac{h^2}{d_1^2}\right)$$

Similarly, for triangle BCD,

$$P_2 = \sqrt{d_2^2 + h^2}$$

$$P_2 = \sqrt{1 + \left(\frac{h}{d_2}\right)^2}$$

$$P_2 = d_2 \left(1 + \frac{1}{2} \frac{h^2}{d_2^2}\right)$$

Thus path difference is,

$$\Delta = (P_1 + P_2) - (d_1 + d_2)$$

$$\Delta = \frac{h^2}{2} \left(\frac{d_1 + d_2}{d_1 d_2}\right)$$

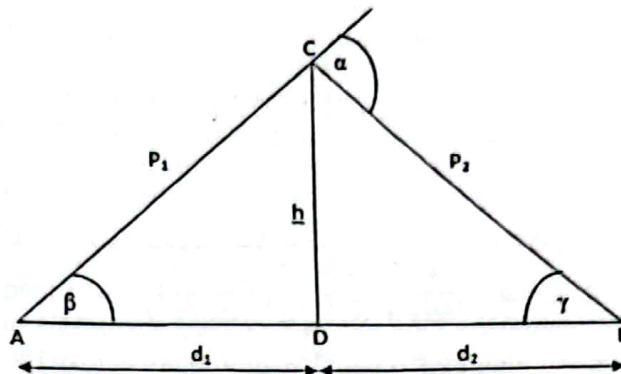
Phase difference is,

$$\varphi = \frac{2\pi}{\lambda} \Delta$$

$$\varphi = \frac{2\pi h^2}{\lambda} \frac{1}{2} \left(\frac{d_1 + d_2}{d_1 d_2}\right)$$

Derivation of Fresnel-Kirchoff diffraction parameter relation

In the following diagram,



$$\tan \beta = \frac{h}{d_1} \approx \beta$$

And,

$$\tan \gamma = \frac{h}{d_2} \approx \gamma$$

From the property of triangles,

$$\alpha = \beta + \gamma$$

Therefore,

$$\alpha = \frac{h}{d_1} + \frac{h}{d_2} = h \left(\frac{d_1 + d_2}{d_1 d_2} \right)$$

From Fresnel-Kirchoff diffraction condition,

$$\varphi = \frac{\pi}{2} v^2$$

Where, v is Fresnel-Kirchoff diffraction parameter,

So,

$$\frac{2\pi h^2}{\lambda} \left(\frac{d_1 + d_2}{d_1 d_2} \right) = \frac{\pi}{2} v^2$$

$$v = h \sqrt{\frac{2}{\lambda} \left(\frac{d_1 + d_2}{d_1 d_2} \right)}$$

Also,

$$\alpha = h \left(\frac{d_1 + d_2}{d_1 d_2} \right)$$

Thus,

$$v = \frac{\alpha(d_1 d_2)}{(d_1 + d_2)} \sqrt{\frac{2}{\lambda} \left(\frac{d_1 + d_2}{d_1 d_2} \right)}$$

$$v = \alpha \sqrt{\left[\frac{(d_1 d_2)}{(d_1 + d_2)} \right]^2 \times \frac{2}{\lambda} \left(\frac{d_1 + d_2}{d_1 d_2} \right)}$$

Hence,

$$v = \alpha \sqrt{\frac{2}{\lambda} \left(\frac{d_1 d_2}{d_1 + d_2} \right)}$$

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Scattering

The actual received power at the receiver is somewhat stronger than claimed by the models of reflection and diffraction. The cause is that the trees, buildings and lampposts scatter energy in all directions. Hence, scattering occurs when an electromagnetic wave falls on an object, which has very small dimensions as compared to the wavelength of the propagating wave. Scattering occurs due to the roughness of the surface upon which EM wave impinges. Roughness is tested by a Rayleigh criterion, which defines a critical height h_c of surface protuberances for a given angle of incidence θ_i , given by,

$$h_c = \frac{\lambda}{8 \sin \theta_i}$$

A surface is smooth if its minimum to maximum protuberance h is less than h_c , and rough if protuberance is greater than h_c .

Practical Link Budget Models

OKUMURA MODEL

The Okumura model is used for Urban Areas is a Radio propagation model that is used for signal prediction. The frequency coverage of this model is in the range of 150 MHz to 1920 MHz and distances of 1 Km to 100 Km. It can be applicable for base station effective antenna heights (h_t) ranging from 30 m to 1000 m. Okumura used extensive measurements of base station-to-mobile signal attenuation throughout Tokyo to develop a set of curves giving median attenuation relative to free space (A_{mu}) of signal propagation in irregular terrain. The empirical pathloss formula of Okumura at distance d parameterized by the carrier frequency f_c is given by

$$L_{50}(\text{dB}) = L_F + A_{mu}(f, d) - G(h_t) - G(h_r) - G_{AREA}$$

where L_F is free space path loss at distance d and carrier frequency f_c , $A_{mu}(f, d)$ is the median attenuation in addition to free-space path loss across all environments, $G(h_t)$ is the base station antenna height gain factor, $G(h_r)$ is the mobile antenna height gain factor, G_{AREA} is the gain due to type of environment. The values of $A_{mu}(f_c, d)$ and G_{AREA} are obtained from Okumura's empirical plots. Okumura derived empirical formulas for $G(h_t)$ and $G(h_r)$ as follows:

$$G(h_t) = 20 \log_{10}(h_t/200), \quad 30\text{m} < h_t < 1000\text{m}$$

$$G(h_r) = 10 \log_{10}(h_r/3), \quad h_r \leq 3\text{m}$$

$$G(h_r) = 20 \log_{10}(h_r/3), \quad 3\text{m} < h_r < 10\text{m}$$

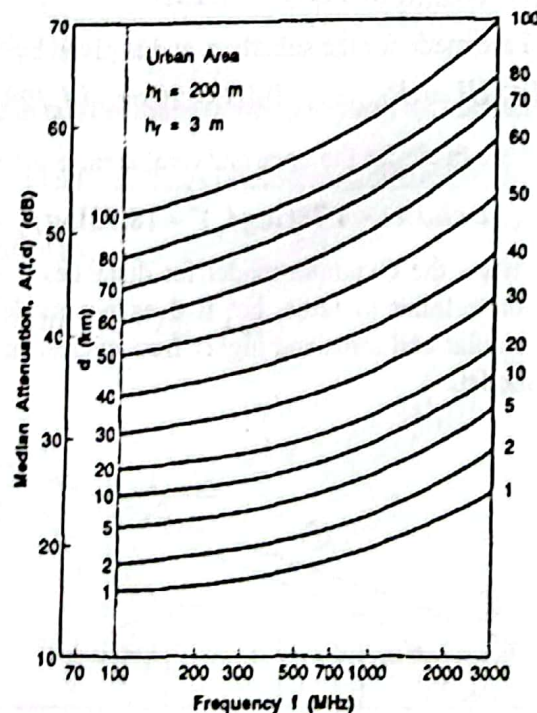


Fig: Okumura Curves

Okumura's model has a 10-14 dB empirical standard deviation between the path loss predicted by the model and the path loss associated with one of the measurements used to develop the model.

HATA MODEL

The Hata model is an empirical formulation of the graphical path-loss data provided by the Okumura and is valid over roughly the same range of frequencies, 150-1500 MHz. This empirical formula simplifies the calculation of path loss because it is closed form formula and it is not based on empirical curves for the different parameters. The standard formula for empirical path loss in urban areas under the Hata model is

$$P_{L,urban}(d)dB = 69.55 + 26.16 \log_{10}(f_c) - 13.82 \log_{10}(h_t) - a(h_r) + (44.9 - 6.55 \log_{10}(h_t)) \log_{10}(d)$$

The parameters in this model are same as in the Okumura model, and $a(h_r)$ is a correction factor for the mobile antenna height based on the size of coverage area. For small to medium sized cities this factor is given by

$$a(h_r) = (1.11 \log_{10}(f_c) - 0.7)h_r - (1.56 \log_{10}(f_c) - 0.8)dB$$

And, for larger cities at a frequencies $f_c > 300$ MHz by

$$a(h_r) = 3.2(\log_{10}(11.75h_r))^2 - 4.97dB$$

Else, it is

$$a(h_r) = 8.29(\log_{10}(1.54h_r))^2 - 1.1dB$$

Corrections to the urban model are made for the suburban, and is given by

$$P_{L,suburban}(d)dB = P_{L,urban}(d)dB - 2(\log_{10}(f_c/28))^2 - 5.4$$

Corrections to the urban model are made for the open and rural area, is given by

$$L_{50}(dB) = L_{50}(urban) - 4.78(\log f_c)^2 - 18.33 \log f_c - 40.98$$

The Hata model well approximates the Okumura model for distances $d > 1$ Km. Hence it is a good model for first generation cellular systems, but it does not model propagation well in current cellular systems with smaller cell sizes and higher frequencies. Indoor environments are also not captured by the Hata model.

Longley-Rice Model

Longley-Rice model is applicable to point to point communication systems in the frequency range from 40 MHz to 100 GHz over different kinds of terrain.

- Median transmission loss predicted through terrain profile and tropospheric refractivity
- Two ray model used to predict received signal level
- Fresnel-kirchoff knife edge model used to predict diffraction losses

Modes of operation

1. Point to point mode: used when detail terrain profile is available
2. Area mode: used when detail terrain profile is not available

Drawback: No correction factor for environmental factors

Walfisch and Bertoni Model

This model considers the impact of rooftops and building heights by using diffraction to predict average signal strength at street level. The model considers the path loss, S , to be a product of three factors,

$$S = P_0 Q^2 P_1$$

P_0 = Freespace path loss

Q^2 = Shadow effect due to tall buildings (rooftop to street diffraction and scatter factor)

P_1 = Diffraction loss due to tall buildings (multiscreen diffraction due to rows of buildings)

In dB, the path loss is given by,

$$S \text{ (dB)} = L_0 + L_{rts} + L_{ms}$$

